

MODULE 03

Shoulder MRI: Rotator Cuff, Instability & Arthroplasty

Goutallier · Patte · CSA · Walch · Seebauer · Hamada · Arthroplasty Complications

1.0 AMA PRA Cat 1

Shoulder

CME Article Review

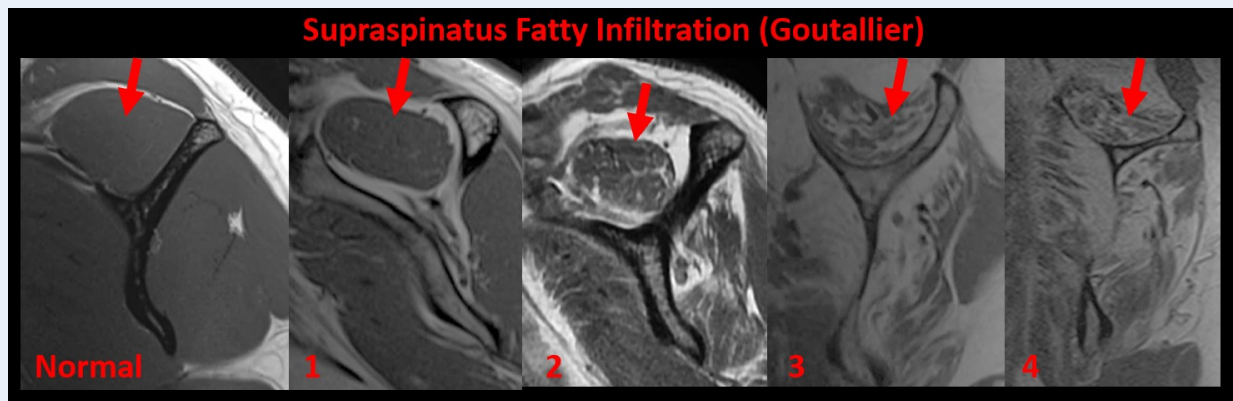
LEARNING OBJECTIVES

1. Apply Goutallier grading (0–4) to rotator cuff muscle fatty infiltration on T1 MRI.
2. Classify supraspinatus retraction using the Patte staging system.
3. Interpret Critical Shoulder Angle (CSA) and its pathoanatomic implications.
4. Classify glenoid morphology using the Walch classification (A1–D).
5. Recognize RCA severity using Seebauer and Hamada classifications.
6. Identify aTSA and rTSA-specific complications: notching, loosening, ALTR.

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1. Rotator Cuff Fatty Infiltration — Goutallier Classification

Fatty infiltration of the rotator cuff musculature is a critical prognostic factor in cuff tear repair: muscle that has undergone substantial fatty replacement does not recover contractile function even after anatomic tendon repair, and high-grade infiltration is a relative contraindication to repair in favor of reverse total shoulder arthroplasty (rTSA) or tendon transfer. The Goutallier classification, originally described on axial CT and now routinely applied to T1-weighted MRI (oblique sagittal images at the level of the scapular spine, the "Y-view"), grades the supraspinatus, infraspinatus, and subscapularis independently on a 5-point scale.



Goutallier grading of supraspinatus fatty infiltration, Grade 0 (normal) through Grade 4, on oblique sagittal T1 MRI.

Goutallier Classification — Muscle Fatty Infiltration

Grade	Description	Clinical Significance
0	Normal muscle, no fatty streaks	Excellent repair candidate
1	Some fatty streaks within muscle	Good repair candidate
2	Fat present but less than muscle (fat < muscle)	Repair generally still favored
3	Fat equal to muscle volume (fat = muscle)	Threshold grade — repair outcomes decline sharply beyond this point
4	Fat exceeds muscle volume (fat > muscle)	Repair unlikely to restore function; consider rTSA or tendon transfer

! Grade ≥ 2 (fat $\geq$ muscle) on any cuff muscle is generally considered the threshold beyond which tendon repair outcomes deteriorate significantly. Always grade all three muscles (supraspinatus, infraspinatus, subscapularis) independently — they frequently differ.

Key References

- Goutallier D, Postel JM, Bernageau J, et al. Fatty muscle degeneration in cuff ruptures: pre- and postoperative evaluation by CT scan. Clin Orthop Relat Res 1994.
- Fuchs B, Weishaupt D, Zanetti M, et al. Fatty degeneration of the muscles of the rotator cuff: assessment by computed tomography versus magnetic resonance imaging. J Shoulder Elbow Surg 1999.

2. Supraspinatus Tendon Retraction — Patte Classification

The Patte classification describes the degree of supraspinatus tendon retraction on coronal MRI, using the humeral head and glenoid as fixed anatomic references. It complements Goutallier grading: retraction predicts whether a tendon can be mobilized back to its anatomic footprint without excessive tension, while Goutallier grading predicts whether the muscle belly itself retains functional capacity. Both are typically reported together in any rotator cuff tear MRI.

Patte Classification — Supraspinatus Retraction (Coronal MRI)

Stage	Tendon Stump Location	Reparability
1	Near its anatomic insertion on the greater tuberosity	Easily reparable
2	Retracted to the level of the humeral head	Reparable, may require mobilization
3	Retracted to the level of the glenoid	Often irreparable primarily; consider tendon transfer or graft augmentation

- Patte stage and Goutallier grade are independent variables that should both be reported — a tendon can be minimally retracted (Patte 1) but have severely fatty muscle (Goutallier 4), or vice versa. Reparability decisions should weigh both.

Key References

- Patte D. Classification of rotator cuff tears. Clin Orthop Relat Res 1990.
- Thomazeau H, Rolland Y, Lucas F, et al. Atrophy of the supraspinatus belly: assessment by MRI in 55 patients with rotator cuff pathology. Acta Orthop Scand 1996.

3. Critical Shoulder Angle (CSA) — Measurement & Pathoanatomic Implications

The Critical Shoulder Angle is measured on a true AP (Grashey) radiograph of the glenohumeral joint, formed by a line connecting the superior and inferior glenoid rim and a second line from the inferior glenoid rim to the lateral-most point of the acromion. CSA captures the combined contribution of glenoid inclination and acromial lateral coverage to glenohumeral joint loading — it is among the strongest predictors available for which shoulders go on to develop rotator cuff disease versus primary glenohumeral osteoarthritis.

Critical Shoulder Angle — Pathoanatomic Correlation

CSA Range	Biomechanical Implication	Associated Pathology
< 30°	Reduced deltoid lever arm; relatively vertical glenoid	Increased risk of primary glenohumeral osteoarthritis
30–35°	Normal range; balanced joint reaction force vector	Low risk of either cuff tear or isolated OA
> 35°	Increased shear/superior translation force on humeral head	Increased risk of rotator cuff tear and accelerated cuff tear arthropathy

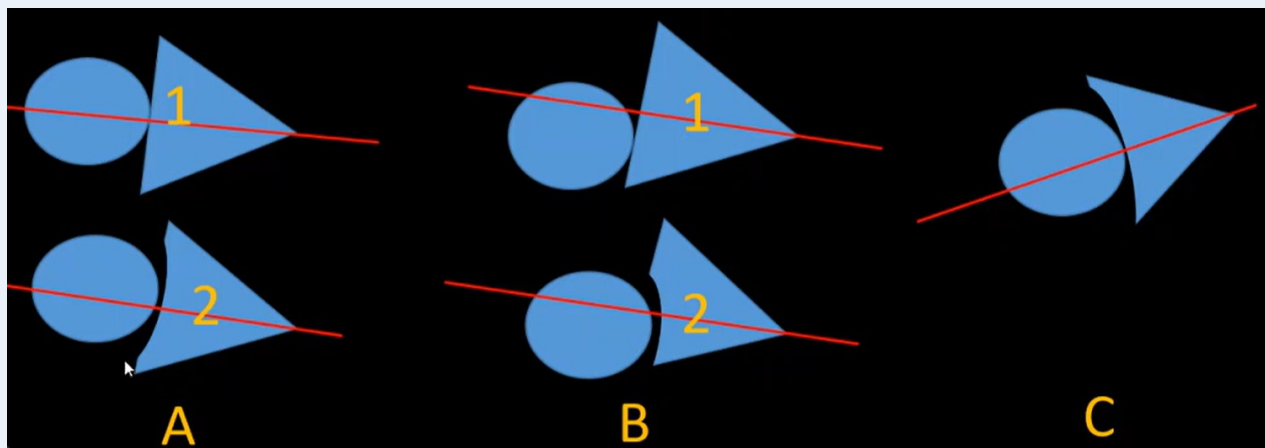
- CSA is a radiograph-based measurement, not an MRI measurement — but every MSK radiologist reading a shoulder MRI for cuff pathology should know whether a prior or concurrent radiograph showed an elevated CSA, since it directly informs surgical planning (e.g. lateralized reverse arthroplasty design in high-CSA patients with cuff tear arthropathy).

Key References

- Moor BK, Bouaicha S, Rothenfluh DA, et al. Is there an association between the individual anatomy of the scapula and the development of rotator cuff tears or osteoarthritis of the glenohumeral joint? Bone Joint J 2013.
- Moor BK, Wieser K, Slankamenac K, et al. Relationship of individual scapular anatomy and degenerative rotator cuff tears. J Shoulder Elbow Surg 2014.

4. Glenoid Morphology — Walch Classification

The Walch classification describes glenoid wear pattern and version on axial CT or MRI at the mid-glenoid level, and is essential preoperative planning information for anatomic versus reverse shoulder arthroplasty. Posterior glenoid wear (Type B) is associated with posterior humeral head subluxation and is a relative contraindication to standard anatomic glenoid component placement without augmentation or version correction.



Walch classification of glenoid morphology: Type A (concentric, A1/A2), Type B (posterior wear/subluxation, B1/B2), Type C (dysplastic, excessive retroversion).

Walch Classification — Glenoid Morphology

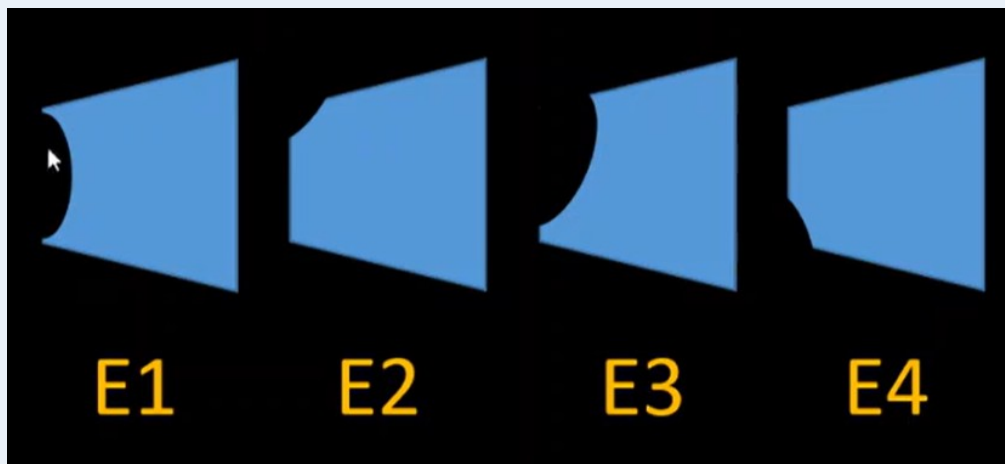
Type	Description	Surgical Implication
A1	Concentric glenoid, minimal erosion, centered humeral head	Standard anatomic component
A2	Concentric glenoid with central erosion	Standard component, possible bone grafting
B1	Posterior joint space narrowing, posterior subluxation, no frank erosion	May need posterior augment or version correction
B2	Biconcave glenoid (posterior wear creates a second articular surface)	High risk of component malposition; reaming or augmented component required
C	Glenoid retroversion > 25° regardless of wear, often dysplastic	Often congenital; complex reconstruction

Key References

- Walch G, Badet R, Boulahia A, Khoury A. Morphologic study of the glenoid in primary glenohumeral osteoarthritis. J Arthroplasty 1999.
- Bercik MJ, Kruse K, Yalozis M, et al. A modification to the Walch classification of the glenoid in primary glenohumeral osteoarthritis using three-dimensional imaging. J Shoulder Elbow Surg 2016.

5. Rotator Cuff Arthropathy — Seebauer & Hamada Classifications

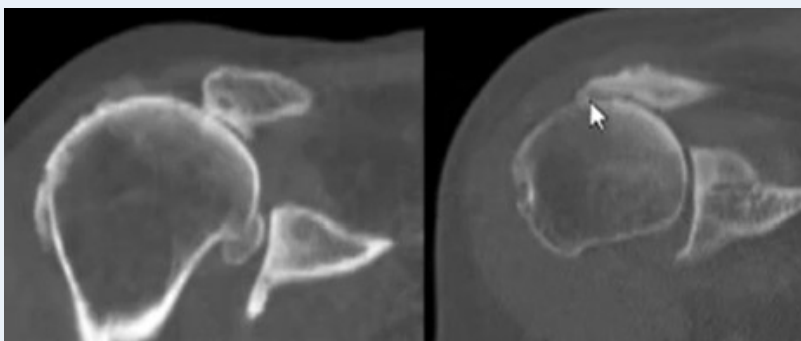
Massive, long-standing rotator cuff tears can progress to rotator cuff arthropathy (RCA): superior humeral head migration, glenohumeral and acromiohumeral degeneration, and eventually a fixed, unstable articulation against the coracoacromial arch. Two complementary classifications are used: Seebauer describes the pattern of glenoid erosion and humeral head centering (informing implant selection), while Hamada grades overall severity based on acromiohumeral distance and acetabularization of the acromion.



Seebauer-type glenoid erosion patterns (E1–E4) seen in rotator cuff arthropathy, used to characterize superior glenoid wear and guide reverse component selection.

Hamada Classification — Rotator Cuff Arthropathy Severity

Grade	Imaging Findings
1	Acromiohumeral distance (AHD) > 6 mm; no acetabularization
2	AHD ≤ 5 mm
3	AHD ≤ 5 mm with acetabularization of the acromion (concave remodeling against humeral head)
4	Glenohumeral arthritic changes present (4A: acetabularization without; 4B: with acromial erosion)
5	Humeral head collapse / osteonecrosis



CT appearance of acetabularization of the acromion in advanced cuff tear arthropathy (Hamada Grade 3–4).

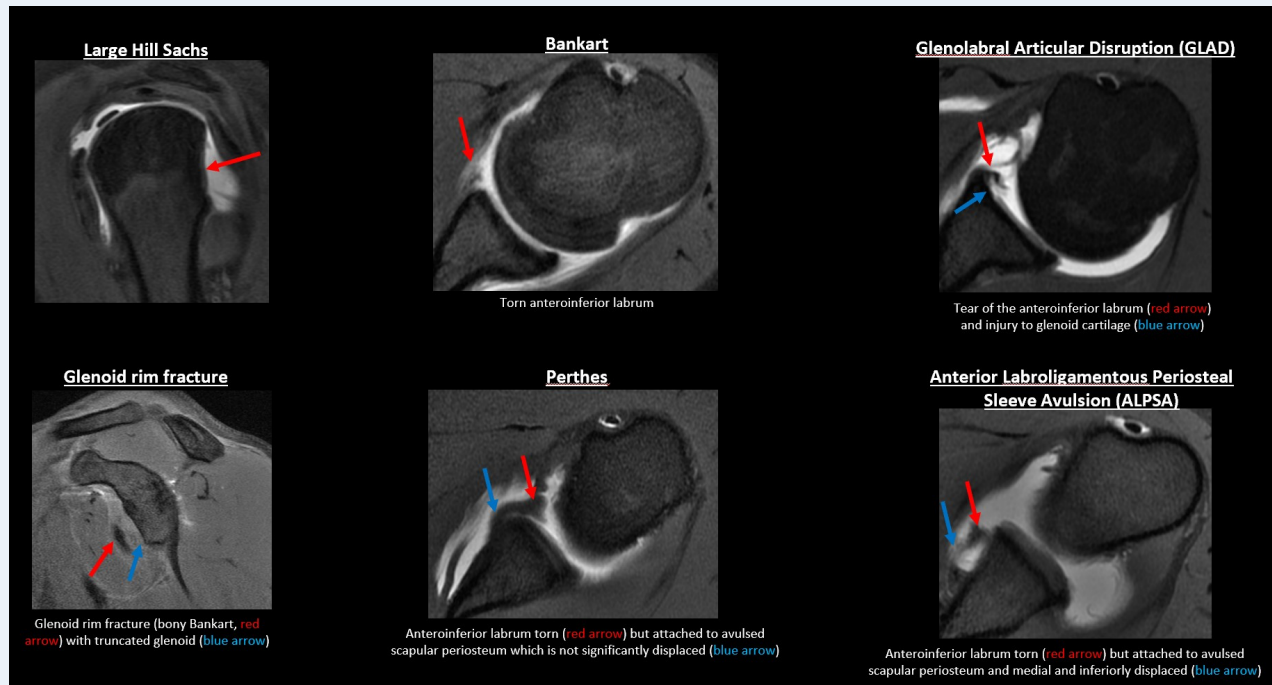
- Seebauer Type 1 (centered humeral head, stable fulcrum) is typically more amenable to a standard reverse component, while Seebauer Type 2 (decentered, superiorly migrated head) often requires a more constrained or lateralized reverse design to restore stability.

Key References

- Hamada K, Fukuda H, Mikasa M, Kobayashi Y. Roentgenographic findings in massive rotator cuff tears: a long-term observation. Clin Orthop Relat Res 1990.
- Seebauer L, Walter W, Keyl W. Reverse total shoulder arthroplasty for the treatment of defect arthropathy. Oper Orthop Traumatol 2005.

6. Glenohumeral Instability — Labroligamentous Injury Spectrum

Anterior glenohumeral instability produces a recognizable spectrum of labroligamentous injuries on MR arthrography, distinguished primarily by the position and displacement of the avulsed anteroinferior labrum relative to the scapular periosteum. Correctly distinguishing these patterns matters because periosteal sleeve injuries (Perthes, ALPSA) are easily missed on routine MRI without arthrographic contrast, since the labrum can appear deceptively close to its normal position.



Spectrum of anteroinferior labral injury: Bankart, ALPSA, GLAD, Perthes, bony Bankart (glenoid rim fracture), and Hill-Sachs lesion.

Anteroinferior Labral Injury Spectrum

Lesion	Key Feature
Bankart	Labrum torn and displaced from glenoid rim, no periosteal attachment preserved
Bony Bankart	Bankart lesion with an associated glenoid rim fracture fragment
Perthes	Labrum torn but periosteum intact and undisplaced — easily missed without arthrography
ALPSA	Labrum torn with periosteal sleeve displaced medially and inferiorly (medialized, healed-down position)
GLAD	Labral tear with an associated glenoid articular cartilage injury, typically a stable labrum
Hill-Sachs	Posterolateral humeral head impaction fracture from anterior dislocation

! ALPSA lesions are a classic source of false-negative instability MRI: the labrum has healed in a medialized position and can mimic a normal-appearing, if slightly redundant, labrum. Always trace the periosteum along the scapular neck on every sequence when instability is the clinical concern.

Key References

- Neviaser TJ. The GLAD lesion: another cause of anterior shoulder pain. Arthroscopy 1993.
- Wischer TK, Bredella MA, Genant HK, et al. Perthes lesion (a variant of the Bankart lesion): MR imaging and MR arthrographic findings with surgical correlation. AJR 2002.

7. Shoulder Arthroplasty Complications — aTSA & rTSA

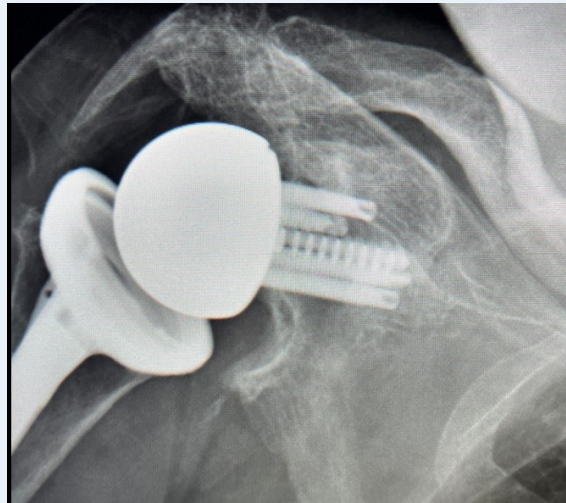
Both anatomic (aTSA) and reverse (rTSA) total shoulder arthroplasty carry distinct, partially overlapping complication profiles. aTSA failure is dominated by glenoid component loosening and rotator cuff failure (since the native cuff must remain functional), while rTSA failure is dominated by scapular notching, instability/dislocation, and acromial or scapular spine stress fracture — a direct consequence of the semi-constrained, medialized or lateralized reverse biomechanics. Both implant types share risk for periprosthetic infection, periprosthetic fracture, and adverse local tissue reaction (ALTR) from metal debris at modular junctions.

Shoulder Arthroplasty Complication Checklist

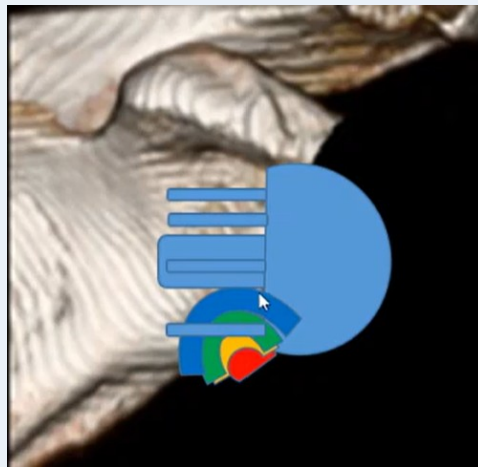
Complication	Typical Implant	Key Imaging Features	Critical?
Scapular notching	rTSA	Erosion of inferior scapular neck from polyethylene/humeral cup impingement; graded Nerot-Sirveaux I–IV	! Yes if Grade III–IV
Glenoid component loosening	aTSA / rTSA baseplate	Progressive radiolucency ≥ 2 mm around pegs/screws; component tilt or migration on serial radiographs	! CRITICAL
Glenoid component dissociation / wear-through	aTSA	Eccentric humeral head position; glenoid bone-on-metal contact; polyethylene wear debris	! YES
Periprosthetic fracture (humeral or scapular)	aTSA / rTSA	Fracture line at or adjacent to stem tip or baseplate screws; assess implant stability	! YES
Instability / dislocation	rTSA (more common than aTSA)	Humeral component displaced from glenosphere; assess soft tissue tension and component version	! YES
Acromial / scapular spine stress fracture	rTSA	Lucent fracture line through acromion or scapular spine, often subtle; correlate with new-onset pain	! YES
Metallosis / ALTR	aTSA / rTSA modular junctions	Periarticular soft tissue mass or synovitis with susceptibility artifact; debris at taper junctions	! YES
Periprosthetic infection	aTSA / rTSA	Periosteal reaction, bone destruction, joint effusion, sinus tract; correlate with labs	! CRITICAL
Rotator cuff failure (subscapularis)	aTSA	Subscapularis discontinuity on axial images; anterior instability risk	Yes (aTSA-specific)

Scapular Notching (rTSA)

Scapular notching results from mechanical impingement of the humeral polyethylene or metal cup against the inferior scapular neck during arm adduction, and is graded by the Nerot-Sirveaux system from Grade 0 (none) to Grade IV (notching extending beneath the baseplate, threatening fixation). Lateralized glenosphere designs reduce but do not eliminate this complication.



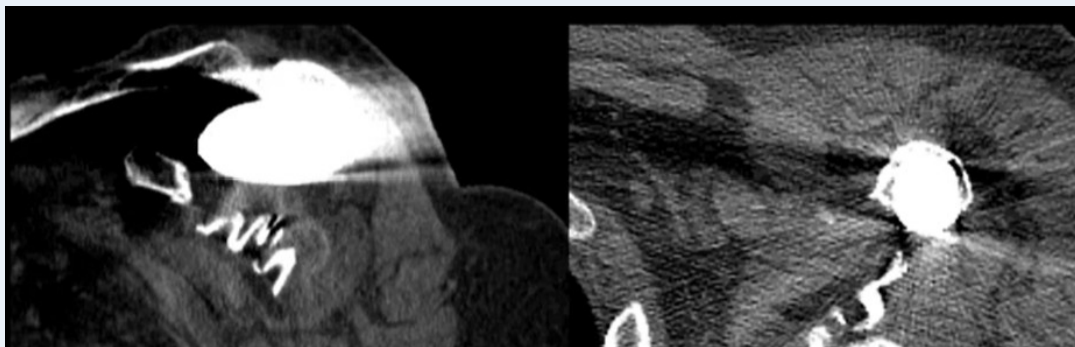
Radiographic scapular notching at the inferior scapular neck following rTSA.



3D CT rendering with color-coded notching severity zones used for grading.

Component Loosening & Failed Anatomic Glenoid

Glenoid component loosening remains the most common reason for aTSA revision. Look for circumferential lucency around the pegs or keel, component tilt compared to baseline, and posterior decentering of the humeral head as an early indirect sign of glenoid wear-through.



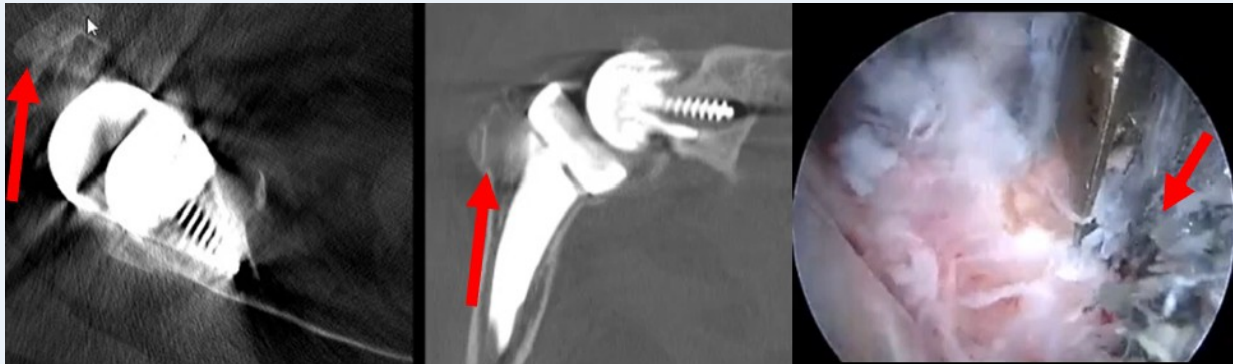
Failed anatomic TSA with glenoid component loosening and posterior decentering of the humeral head on axial CT.



rTSA baseplate loosening with peri-screw lucency on CT.

Metallosis & Adverse Local Tissue Reaction

Although less common than at the hip, modular shoulder arthroplasty components can generate metal debris at the head-neck or glenosphere-baseplate junction, producing periarticular synovitis or a pseudotumor-like soft tissue mass with susceptibility artifact on MRI.



Metallosis at rTSA with periarticular soft tissue reaction; intraoperative correlate shown at right.

Periprosthetic Infection

Periprosthetic joint infection should be considered whenever there is rapid-onset pain, new periosteal reaction, aggressive-appearing bone destruction, or a joint/soft tissue collection — particularly in the absence of mechanical loosening to otherwise explain the findings.



Periprosthetic infection following TSA with extensive periosteal reaction and bone destruction along the humeral stem.

- When dictating a postoperative shoulder arthroplasty MRI or CT, systematically address: component position and version, peri-component lucency/loosening, notching grade (if reverse), cuff integrity (if anatomic), periprosthetic fracture, and any soft tissue mass or collection suggesting infection or ALTR — these should all be itemized even when the study is for a different primary indication.

Key References

- Nerot C, Sirveaux F. Grammont reverse prosthesis. In: Shoulder Arthroplasty. Springer 2007.
- Sanchez-Sotelo J. Total shoulder arthroplasty. Open Orthop J 2011.
- Cheung E, Willis M, Walker M, et al. Complications in reverse total shoulder arthroplasty. J Am Acad Orthop Surg 2011.